

Module – 1 : Understanding of Hardware and Its Components

Section 1 : Multiple Choice

1. Which of the following is NOT a component of the CPU ?

Ans : RAM

2. What is the function of RAM in a computer ?

Ans : To temporarily store data and instructions that the CPU is currently using.

3. Which of the following is a primary storage device ?

Ans : 1 and 2 both

4. What is the purpose of a GPU ?

Ans : Process and display graphics, images, and video on the screen.

Section 2 : True or False

5. The motherboard is the main circuit board of a computer where other components are attached.

Ans : True

6. A UPS (Uninterruptible Power Supply) is a hardware device that provides emergency power to a load when the input power source fails.

Ans : True

7. An expansion card is a circuit board that enhances the functionality of a component.

Ans : True

Section 3 : Short Answer

8. Explain the difference between HDD and SSD

Ans :

HDD (Hard Disk Drive)	SSD (Solid State Drive)
Uses magnetic storage with spinning disks	Uses flash memory chips (no moving parts)
Slower read/write speed	Much faster read/write speed
More storage at lower cost	More expensive
Produces noise and heat	Silent and cool
Less durable	More durable

9. Describe the function of BIOS in a computer system.

Ans : BIOS (Basic Input/Output System) is software stored on the motherboard. Its function is to **initialize and test hardware components** when the computer starts and then **load the operating system** into memory.

10. List and briefly explain three input devices commonly used with computers

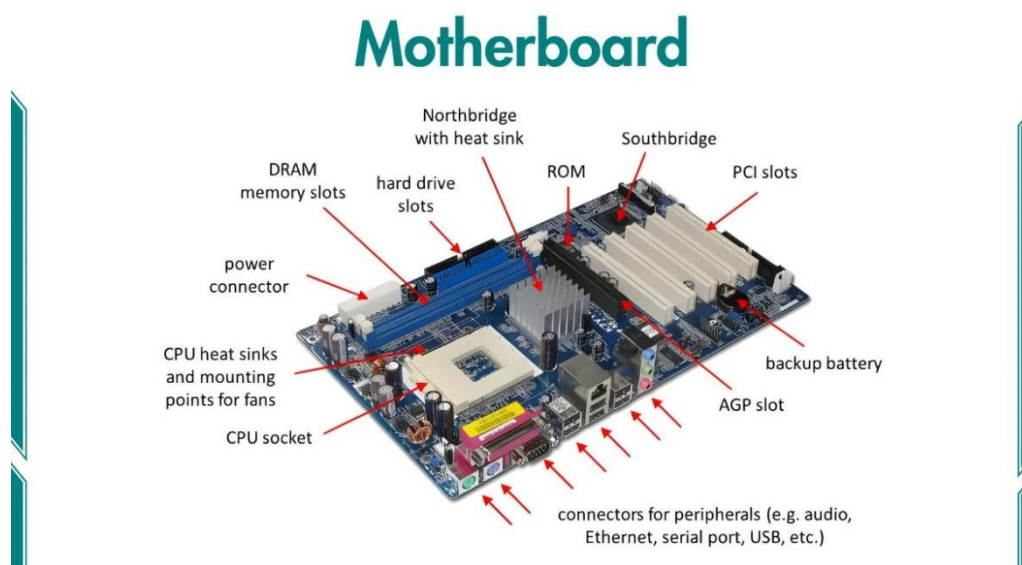
Ans :

1. **Keyboard** – Used to enter text, numbers, and commands into the computer.
2. **Mouse** – Used to control the pointer and select objects on the screen.
3. **Scanner** – Used to convert physical documents or images into digital format.

Section 4 : Practical Application

11. Identify and label the following components on a diagram of a motherboard :

Ans :



a) CPU :

- Big **square socket**.
- Usually near the **top middle** of the motherboard.

b) RAM slots :

- **Long thin slots** next to the CPU.
- Usually **2 or 4 slots** for memory sticks.

c) SATA connectors :

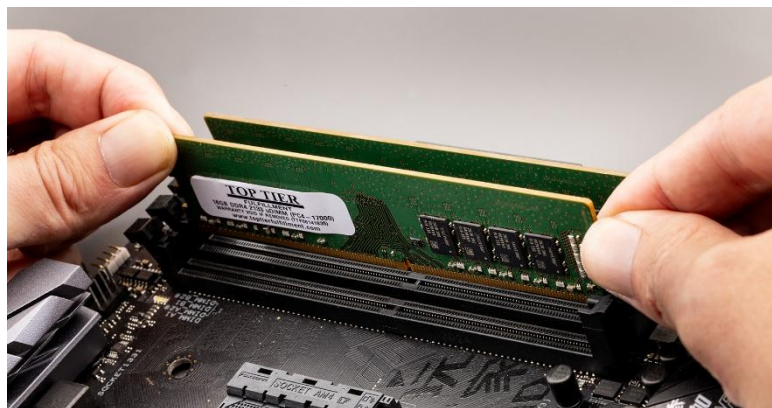
- **Small L-shaped ports** on the right side.
- Used for **HDD/SSD** connections.

d) PCI-E slot :

- **Long slot** near the bottom.
- Used for **graphics card**.

12. Demonstrate how to install a RAM module into a computer.

Ans :



Step 1: Power Off the Computer

- Shut down the PC completely.

Step 2 : Open the Computer Case

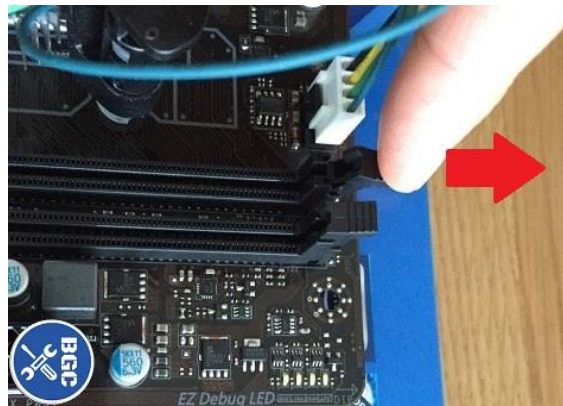
- Slide the panel off to access the motherboard.

Step 3 : Locate the RAM Slots

- Look for the **long, thin slots** next to the CPU.
- These are called **DIMM slots**.

Step 4 : Unlock the

- Each RAM slot has **two small clips** on the ends.
- Gently push them **outward** to open.



Step 5 : Align the RAM Module

- Look at the **notch** on the RAM stick.
- Match it with the ridge inside the slot.
- The RAM will only fit **one way**.



Step 6 : Insert the RAM Module

- Hold the RAM by the edges.
- Place it into the slot.
- Press down **firmly and evenly** on both ends.

Step 7 : Close the Case and Power On

- Put the side panel back.
- Connect the power cable.
- Turn on the computer.
- The system should now detect the new RAM.

Section 5 : Essay

13. Discuss the importance of proper cooling mechanisms in a computer system. Include examples of cooling methods and their effectiveness.

Ans :

- Proper cooling is very important in a computer because every component, like the **CPU, GPU, and power supply**, produces heat when the computer is running. If this heat is not controlled, the computer can slow down, freeze, or even get damaged.

➤ Why Cooling Is Important

1. Prevents Overheating

- Cooling keeps the temperature low so the computer does not get too hot.

2. Increases Performance

- When a computer stays cool, the CPU and GPU can work faster without slowing down.

3. Improves Life of Components

- Less heat means parts last longer and do not wear out quickly.

4. Prevents System Shutdowns

- Computers may shut down automatically if they get too hot. Cooling helps avoid this problem.

➤ Examples of Cooling Methods

1. Air Cooling (Most Common and Simple)



- Uses **fans** and **heat sinks**.
- Fans push hot air out and bring cool air in.
- It is **cheap, easy to install, and effective** for normal computers.
- **Effectiveness :**
Good for everyday use, office work, gaming at moderate levels.

2. Liquid Cooling (Water Cooling)



- Uses **liquid** to move heat away from the CPU or GPU.
- Includes a pump, tubes, water block, and radiator.
- **Effectiveness:**
Very good cooling, quieter than fans, best for **gaming PCs** or **high-performance systems**.

3. Case Ventilation



- Good airflow inside the computer case.
- Fans should bring cool air in from the front and push hot air out from the back or top.
- **Effectiveness:**
Improves overall temperature and prevents heat buildup.

14. Explain the concept of bus width and its significance in computer architecture.

Ans :

- A bus in a computer is like a highway that carries data between different parts of the computer (CPU, memory, and other components).

➤ What is Bus Width ?

- **Bus width** is the **number of bits** a bus can carry at one time.
- It is usually measured in **bits** (8-bit, 16-bit, 32-bit, 64-bit, etc.).

➤ Why Is Bus Width Important ?

1. More Bus Width = More Data Transfer

- A wider bus can move more data at the same time, just like a wider road allows more cars to travel.

2. Affects Overall Speed of the Computer

- If the bus is narrow, the CPU has to wait longer for data to arrive.
- A wider bus reduces waiting time, making the computer run faster.

3. Important for Modern CPUs and RAM

- Modern computers use **64-bit buses** because they need to move large amounts of data quickly, especially for :
 - Gaming
 - Video editing
 - Multitasking

4. Helps Support Larger Memory

- A wider bus allows the CPU to access larger amounts of RAM efficiently.

- **Bus width** = how many bits can travel at once.
- **Wider bus** = **faster data transfer** and **better performance**.
- **Modern computers** use **64-bit bus width** for high speed and efficiency.